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## **Preventing 25% Production Loss: Combating Hydrogen Induced Cracking (HIC) Through Equipment Derating**

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### **Abstract**

Hydrogen-Induced Cracking (HIC) poses a significant challenge to plant integrity, endangering the reliability and safety of pivotal process equipment. At one of our oil processing plants, HIC was discovered in one of the four stripping columns, prompting urgent measures to protect production and uphold health, safety, and environmental (HSE) standards. This paper outlines a comprehensive investigation into the HIC issue affecting the stripping column, utilizing a robust methodology, which includes metallurgical analysis, process data evaluation, and an extensive risk assessment. Multiple alternatives were explored to maintain operational continuity, such as pressure derating,

Pressure Safety Valve (PSV) replacement, sectional repair, and full column replacement. Ultimately, pressure derating and PSV upgrades were adopted as swift solutions, ensuring safe operation with minimal disruption to production and export commitments. This paper emphasizes the link between process conditions and HIC occurrence, the crucial role of non-destructive testing (NDT) for early detection, and the significance of systematic risk analysis.

Recommendations are offered for industry practitioners to proactively manage HIC-related risks in similar process settings.

### **Introduction**

Hydrogen-induced cracking is a form of material degradation arising from exposure to hydrogen sulfide (H<sub>2</sub>S) in a wet H<sub>2</sub>S environment or sour service. It results from hydrogen accumulation at steel imperfections, such as laminations or inclusions. Atomic hydrogen produced by wet H<sub>2</sub>S or a hydrofluoric environment combines at these imperfections to form hydrogen molecules that cannot diffuse through the steel. This accumulation leads to high pressure buildup, causing local stresses that exceed the material's yield strength near these imperfections. The yielding and subsequent plastic deformation manifest as bulging and result in blisters. This originates from the formation and expansion of small, crack-like defects within the steel, often appearing as tiny internal gaps or blisters that are predominantly laminar, running parallel to both the interior and exterior surfaces of the steel.

Once hydrogen blistering starts at a location, it can grow and further develop into Hydrogen Induced Cracking/Stress Oriented Hydrogen Induced Cracking (SOHIC)/Sulfide Stress Corrosion Cracking.

The phenomenon is marked by crack formation within the steel matrix due to hydrogen atom ingress and accumulation, which can lead to rapid equipment failure and compromise plant safety. HIC is especially problematic in areas subjected to high-pressure operations and corrosive process conditions, where the risk of hydrogen ingress is elevated. HIC is most commonly observed when hydrogen, generated in aqueous H<sub>2</sub>S hydrocarbon settings, diffuses into the steel substrate. Even at relatively low temperatures, atomic hydrogen produced from corrosion reactions involving wet H<sub>2</sub>S can penetrate the steel and accumulate at sites of inclusions or impurities.

Typically, hydrogen atoms would recombine at the metal's surface and escape as gas, but the presence of H<sub>2</sub>S hinders this recombination process. Instead, hydrogen is driven into the steel's interior, where it collects and leads to structural degradation.

Regarding its impact, HIC is often minor and does not immediately compromise material integrity unless it becomes widespread. When cracks proliferate, they can interact and join due to increased internal pressure or local stress concentrations, sometimes advancing toward welds or moving through the steel thickness in a pattern known as "stepwise" cracking.

One of the oil plants at ADNOC Offshore, a strategic facility in the region's oil and gas infrastructure, relies on four stripping columns for removing impurities from hydrocarbon streams, a process vital for maintaining product quality and compliance with export standards. The detection of HIC in one of these columns presented a significant integrity challenge, necessitating a carefully balanced response to maintain production, avoid costly downtime, and adhere strictly to HSE protocols.

This paper presents a detailed account of the investigation, evaluation, and mitigation strategies undertaken to address the HIC issue in the stripping column. By integrating metallurgical expertise, operational insights, and advanced safety analysis, the team developed and implemented effective solutions that preserved plant reliability and minimized production losses. The findings contribute to a broader understanding of HIC management in process equipment and offer practical guidance for similar scenarios in the industry.

## **INSPECTION TECHNIQUES APPLIED**

### **Automated Mapping**

- To perform high-speed ultrasonic C-scan on the shell/plate of a COLD STRIPPING COLUMN, a fully automated two-axis raster scanning concept was utilized. For the AUT HIC inspection, a triplex probe was used, which contains two outward-facing 45-degree shear wave transducers and a dual 0-degree transducer in the center. The triplex probe technology is particularly well-suited for HIC inspection.
- A 5 × 5 mm index resolution was used to achieve good image resolution for data acquisition. At this index, data was acquired in 5 mm in the line scan.
- A 1 × 1 mm index resolution was used in some areas where detailed size indication is required.

### **Automated Phased Array Technology, Time of Flight Diffraction (ToFD) & Total Focusing Method (TFM)**

- Automated Phased Array & Total Focusing Method techniques were applied where only one side is accessible, such as a nozzle or circumferential weld near an obstacle area.

- Time of flight diffraction was applied on the weld or near the weld where both sides are accessible, such as circumferential welds and longitudinal welds. ToFD was used for the initial screening of HIC Detection for the weld.
- Further to the above steps, the suspected areas were inspected with Time of Flight Diffraction, Automated Phased Array + 30-sector scan, and TFM. The purpose was to characterize indications/accurate depth measurements and verify the presence of any stepwise cracks.
- APAUT, TOFD, and TFM were used for verification of HIC.

## Evaluation

The initial detection of HIC in the stripping column was facilitated by routine non-destructive testing (NDT) as part of the plant's preventive maintenance schedule. Ultrasonic testing revealed internal cracking in the lower section of the column, raising immediate concerns about the equipment's structural integrity and operational safety.

A multidisciplinary team was mobilized to conduct a comprehensive evaluation, encompassing the following key activities:

- Automated ultrasonic testing was performed on the cold stripping column. Based on the previous periodic assessment, the company identified areas of the vessel that included lamination, inclusions, blisters, and HIC defects.
- Furthermore, the necessity of comprehensive coverage, encompassing all shell courses on the vessel that are considered potentially susceptible to hydrogen-induced crack damage, was highlighted.
- Specific locations were identified to reconfirm the existence of the identified cracks in cold stripping column.
- The UAUT system represents a cutting-edge advancement in non-destructive testing, offering extensive customization options. It leverages advanced techniques such as Ultrafast AUT (Corrosion Mapping), Automated Phased Array, Total Focusing Method (TFM), and Time of Flight Diffraction (ToFD), making it suitable for a wide range of applications, including vessels, column, and piping inspections.
- Process Data Analysis: Plant operational data covering temperature, pressure, feed composition, and column performance over several months were reviewed.
- Risk Assessment: The team carried out a detailed risk analysis using Process Hazard Analysis (PHA) workshops. The assessment focused on the likelihood of column failure, the potential impact on production and exports, and HSE implications. Risk matrices highlighted the severity of HIC progression and the urgency of mitigation to avoid catastrophic failure.
- Equipment Alternatives Evaluation: Several mitigation alternatives were considered: derating the column pressure and changing the PSV set point, PSV replacement, repairing the damaged column section, and full replacement of the column. Each option was evaluated against criteria of implementation speed, cost, production impact, and HSE compliance.

## HSE Risks

Hydrogen-Induced Cracking poses substantial HSE risks due to its potential to cause sudden and catastrophic equipment failure. The primary hazards associated with HSE in stripping columns include:

- Personnel Safety: A compromised column could result in leaks or rupture, exposing personnel to hazardous hydrocarbons, hydrogen gas, and debris. The risk of fire, explosion, and toxic exposure is elevated, particularly given the high-pressure and flammable nature of process streams.

- **Environmental Impact:** Uncontrolled release of hydrocarbons or byproducts can lead to significant environmental contamination, affecting local habitats and marine ecosystems surrounding the process plant. Regulatory compliance and environmental stewardship are critical aspects under scrutiny.
- **Process Safety:** The integrity of the stripping column is integral to safe plant operation. HIC progression increases the likelihood of process upsets, unplanned shutdowns, and emergency intervention, heightening overall operational risk.
- **Business Continuity:** Production loss due to equipment failure impacts export commitments and financial performance. The need to balance rapid remediation with long-term integrity drives decision-making under tight operational constraints.

To mitigate these risks, the plant implemented enhanced monitoring protocols, emergency preparedness measures, and strict adherence to HSE best practices during inspection. The selection of mitigation strategies was governed by a commitment to minimizing personnel exposure, environmental harm, and process disruption.

## Result

The comprehensive evaluation led to the following outcomes:

- **Column Derating:** The integrity assessment necessitated the reduction of column operating pressure from the original design pressure of 30 psig to 15 psi. This derating was vital to prevent exacerbation of cracking and ensure continued safe operation.
- **PSV Assessment and Upgrade:** The existing Pressure Safety Valve (PSV), calibrated for 30 psi, was rendered unsuitable by the new operating conditions. Replacement of the PSV spring was identified as the preferred solution, requiring 9–13 weeks for completion due to lead times on specialized parts. Installation of an alternative PSV, already available on-site and set for 15 psi, was selected as the quickest and safest option following thorough Process Hazard Analysis workshops.
- **Equipment Integrity Safeguards:** Regular inspection intervals were instituted, focusing on HIC-prone locations, supported by advanced NDT techniques, including phased array ultrasonics. The enhanced inspection regimen enabled early detection of HIC progression and informed maintenance planning.
- **Operational Continuity:** By selecting pressure derating and rapid PSV replacement, the team avoided a projected 25% reduction in production that would have jeopardized export obligations and plant profitability. The solution maintained safe operation within the revised pressure envelope and deferred the need for full column replacement, which would have involved extensive downtime.

## Discussion

The case of HIC in the stripping column offers several important insights into the management of process integrity challenges in oil and gas facilities:

- **Plan, developed for repair and replacement, de-rating along with continuous monitoring is implemented for safe operation to avoid production impact.**
- **Effectiveness of Advanced NDT Techniques:** Early detection through sophisticated NDT methods is essential for identifying HIC before it compromises equipment integrity. Techniques such as phased array ultrasonics and magnetic flux leakage scanning provide high-resolution imaging, enabling precise localization and sizing of internal cracks. Routine implementation of these methods should be considered standard practice in high-risk environments.

- **Risk-Based Decision Making:** The selection of mitigation strategies was driven by a robust risk assessment framework, balancing operational needs against safety and environmental considerations. Derating and PSV replacement were chosen for their expedience, limited impact on production, and compliance with risk reduction targets. Alternative options, such as sectional repair or full column replacement, were ruled out due to longer implementation times and higher associated risks.
- **Importance of Proactive Maintenance:** The necessity of regular inspection and timely intervention cannot be overstated. HIC can progress rapidly under unfavorable conditions, making reactive maintenance insufficient. Proactive programs that combine scheduled NDT, condition monitoring, and predictive analytics are essential for sustaining equipment reliability.
- **HSE Integration:** All phases of the investigation and remediation were executed under strict HSE protocols, with continuous assessment of personnel safety, environmental impact, and process hazards. The integration of HSE considerations into technical decision-making is a model for best practice in the industry.

## Production Loss Prevention

The shutdown of a single stripping column can result in a significant reduction of the total plant production capacity by 25%, leading to substantial production losses. The conventional approach to rectifying this issue involves replacing the Pressure Safety Valve (PSV) with a new one, as recommended in the derating report. However, this method is time-consuming, with equipment repair and replacement taking over 9-12 months.

An additional risk is the potential for the integrity assessment of the other stripping column to yield a similar result of zero remaining life. This necessitates the exploration of alternative solutions to address the issue and prepare for potential future failures of other stripping columns.

Derating the column from 30 Psig to 15 Psig requires the replacement of the existing PSVs, as merely replacing the spring would only reduce the pressure to 26 Psig. Consequently, a new PSV is essential. The manufacturing and supply of a new PSV with a 15 Psig set point would take approximately 9–14 weeks, significantly impacting production.

To ensure business continuity, it is imperative to secure PSVs as soon as possible. The company initiated a search for a PSV with the same characteristics but a lower set point of 15 Psig. An inventory check revealed several pieces of equipment equipped with spare PSVs. The technical team identified several standby PSVs suitable for the stripping column. The necessary Process Hazard Analysis (PHA) workshops and sessions were conducted to ensure that all credible scenarios were assessed, evaluated, and remained within acceptable limits without compromising Health, Safety, and

Environment (HSE) standards when installing the new PSV.

## Recommendation

The following recommendations are provided for practitioners managing similar HIC challenges in process equipment:

- **Implement Comprehensive Inspection Regimes:** Utilize advanced NDT technologies for frequent, targeted inspection of critical equipment, especially in hydrogen-rich environments. Early identification of HIC enables timely intervention and reduces the risk of catastrophic failure.
- **Control Process Parameters:** Maintain strict control over operating pressure, temperature, and feed composition to minimize conditions conducive to hydrogen ingress. Automated monitoring and alarm systems should be employed to detect deviations and trigger corrective action.



- **Risk-Based Maintenance Planning:** Adopt risk-based maintenance strategies that prioritize inspection and remediation resources according to equipment vulnerability and process criticality. Integrate FMEA and PHA into maintenance planning to guide decision-making under uncertainty.
- **Rapid Mitigation Solutions:** Prepare for contingency scenarios by prequalifying alternative equipment, such as PSVs with adjustable settings or spare units calibrated for reduced pressure operation. Expedite procurement and installation processes to minimize downtime.
- **HSE Leadership:** Maintain unwavering commitment to HSE standards throughout inspection, maintenance, and operational activities. Regularly update emergency preparedness plans and conduct personnel training focused on HIC-related hazards.
- **Continuous Improvement:** Document lessons learned from HIC events and integrate them into operational procedures, inspection protocols, and design standards. Disseminate findings to promote industry-wide advancement in integrity management.
- Periodic inspections are then scheduled at intervals determined by the severity and progression risk of the defects, as well as operational and environmental conditions. During each inspection, the same areas are re-examined to detect any changes in the size or characteristics of the defects.

## Conclusion

Hydrogen-Induced Cracking (HIC) remains a significant threat to process equipment integrity, particularly in hydrogen-rich and high-pressure environments such as stripping columns. Through a comprehensive investigation combining process data evaluation, and risk assessment, the technical team identified and implemented effective solutions—pressure derating and PSV replacement—that preserved operational continuity and minimized production losses. The integration of advanced NDT, proactive maintenance, and rigorous HSE protocols proved essential to effective risk management.

This paper underscores the importance of early detection technologies, and the necessity of risk-based, HSE-focused decision-making. The recommendations provided herein offer a practical roadmap for industry practitioners confronting HIC-related challenges, promoting safer, more reliable, and environmentally responsible operations in complex process environments. Continuous vigilance and improvement are imperative to sustaining long-term equipment integrity and operational excellence.

In addition to these key points, it is essential to recognize the evolving nature of industrial processes and the continuous advancements in technology. As new materials and detection methods emerge, industry practitioners must remain adaptable and proactive in their approach to HIC management. This includes investing in ongoing training and development for personnel, as well as fostering a culture of safety and innovation within organizations.

Furthermore, collaboration and knowledge-sharing among industry stakeholders can significantly enhance the effectiveness of HIC mitigation strategies. By leveraging collective expertise and experiences, companies can develop more robust and comprehensive solutions to address the multifaceted challenges posed by HIC.

Ultimately, the successful implementation of the recommendations outlined in this paper will not only enhance operational safety and reliability but also contribute to the broader goals of sustainability and environmental stewardship. By prioritizing HIC management and continuously striving for improvement, industry practitioners can ensure the long-term integrity and success of their operations.

## Abbreviations

|            |                                 |
|------------|---------------------------------|
| <b>HIC</b> | Hydrogen-Induced Cracking       |
| <b>HSE</b> | Health, Safety, and Environment |
| <b>NDT</b> | Non-Destructive Testing         |
| <b>API</b> | American Petroleum Institute    |

|             |                                                |
|-------------|------------------------------------------------|
| <b>ASME</b> | American Society of Mechanical Engineers       |
| <b>NACE</b> | National Association of Corrosion Engineers    |
| <b>ISO</b>  | International Organization for Standardization |
| <b>ASTM</b> | American Society for Testing and Materials     |
| <b>FMEA</b> | Failure Modes and Effects Analysis             |
| <b>RBI</b>  | Risk-Based Inspection                          |
| <b>HIC</b>  | Hydrogen-Induced Cracking                      |
| <b>HSE</b>  | Health, Safety, and Environment                |
| <b>NDT</b>  | Non-Destructive Testing                        |
| <b>API</b>  | American Petroleum Institute                   |
| <b>PHA</b>  | Process Hazard Analysis                        |
| <b>PSV</b>  | Pressure Safety Valve                          |

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